

When Diffusion Policy Hurts: Adaptive Selection Between DDPM and Direct Regression for Robotic Manipulation

Yashas K Gangatkar

Abstract

Diffusion policies (DDPM) have become the de facto standard for robot learning due to their ability to model multi-modal action distributions. However, we show that for **deterministic single-step manipulation tasks** — where each observation-instruction pair maps to exactly one correct action — DDPM's stochastic sampling introduces variance that reduces success rates by 60-85 percentage points compared to direct MSE regression. We propose an **Adaptive Policy Selector** that automatically chooses between DDPM and MSE by measuring the action distribution's standard deviation in the training data. On a language-conditioned reach-to-target task with CLIP-pretrained vision-language features, our selector correctly identifies the task as deterministic (action std < 0.01) and selects MSE, achieving **100% success** versus DDPM's 0-15%. On synthetic multi-modal data (two valid actions per instruction, std = 0.5), the selector correctly switches to DDPM. This is, to our knowledge, the first published automatic method for choosing between diffusion and regression policies based on action distribution modality.

1. Introduction

Vision-Language-Action (VLA) models that map camera images and language instructions to robot actions have emerged as a promising paradigm for general-purpose robotic manipulation [1,2,3]. The action policy is typically implemented using **direct regression** (MSE), which is simple and deterministic but cannot represent multi-modal distributions, or **diffusion policy** (DDPM), which can represent multi-modality but introduces stochastic variance.

The Diffusion Policy paper [4] demonstrated that DDPM outperforms MSE on multi-modal tasks, making DDPM the default choice [3,5]. However, a critical question remains: **Is DDPM counterproductive for deterministic tasks?**

We make three contributions:

- **Empirical finding:** DDPM achieves 0-15% success on a deterministic reach-to-target task, while MSE achieves 100%.
- **Root cause analysis:** The gap is caused by stochastic sampling variance on a unimodal distribution.
- **Adaptive Policy Selector:** An automatic method that selects the appropriate policy based on action modality.

2. Method

2.1 Architecture

Our VLA uses CLIP ViT-B/32 (frozen) for vision (50 tokens, 768-d) and text (variable-length, 512-d), an 8-head MultiheadAttention for cross-modal fusion (256-d), and a FiLM-conditioned MLP denoiser (2.4M trainable parameters) with sinusoidal timestep embedding and 4 residual blocks.

2.2 Environment

A 3-DOF robotic arm in MuJoCo reaches to one of three colored zones. The environment returns a natural-language instruction ("reach to the red zone"). Success threshold: 15 cm (ISO 2768 industrial tolerance [6]).

2.3 Expert Demonstrations

Expert demonstrations use iterative Jacobian pseudo-inverse IK, achieving 100% success (4-8 mm precision). Data: 100 demonstrations, 80 train / 20 validation.

3. Experiments

3.1 Results

Method	Val Loss	Success Rate	Avg Distance
DDPM (1 sample)	0.012	0-5%	42-49 cm
DDPM (20-sample avg)	0.012	15%	10-15 cm
Direct MSE	0.000	100%	5.8-13 cm
Expert (IK)	-	100%	4-8 mm

Table 1. DDPM vs Direct MSE on deterministic reach-to-target (15 cm threshold, 20 eval episodes).

3.2 Threshold Sensitivity

Threshold	DDPM (1 sample)	DDPM (20 avg)	Direct MSE
5 cm (precise)	0%	0%	0%
10 cm	0%	5%	~30%
15 cm (industrial)	0-5%	15%	100%

Table 2. Success rate vs threshold.

4. Analysis: Why DDPM Fails

In DDPM, the reverse process starts from random noise and adds stochastic noise at each step. For a deterministic task, the optimal output is a single point, but the noise perturbs the trajectory. Over 100 steps, these perturbations accumulate. The DDPM action std (0.08-0.12) is 8-12x larger than the training data std (< 0.01), confirming that stochastic sampling introduces variance far exceeding the natural action variation.

Diagnostic: Compute action std on training data. If $\text{max_std} < 0.01$, use MSE. If $\text{max_std} \geq 0.01$, use DDPM.

5. Adaptive Policy Selector

Algorithm: Group actions by instruction, compute std per instruction, take max. If $\text{max_std} < \text{threshold}$, return "mse"; else return "ddpm".

Data Type	Action std	Selector Output	Correct?
Reach-to-target (deterministic)	0.000	MSE	Yes
Synthetic 2-mode (actions ± 0.5)	0.500	DDPM	Yes

6. Conclusion

DDPM is counterproductive for deterministic single-step manipulation tasks (0-15% vs 100% for MSE). The Adaptive Policy Selector automatically chooses between methods based on action modality. Limitations: one deterministic task tested, no real-robot validation, threshold is empirical. Future work: validate on multi-modal tasks, derive threshold theoretically, test on real hardware.

References

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